Statistical Appendix

Companion to "Eleven Language Models in a Narrow Band of a Moral-Judgment Instrument." Every number here regenerates from the raw runs with one stdlib script, `analysis/build_appendix.py` (random seed 20260720; the A5 model-resampling draws use their own seed, 20260910), except the re-run-averaging figures in A3, which `validity/aggregation_artifact.py` computes from this appendix's output. Tables first; the report carries the prose.

A1. Sample and data

Eleven models from nine labs (three Chinese: DeepSeek, MiniMax, Moonshot), called by the identifiers below on 2026-07-18 and 2026-07-19. Two further models completed the unframed cell but not the framing grid, Cohere Command A (three of eight framings) and Google Gemini 3 Pro (one), and are outside the panel; they appear in A8 only.

key	provider	model identifier
opus	Anthropic	`claude-opus-4-8`
sonnet	Anthropic	`claude-sonnet-5`
gpt55	OpenAI	`gpt-5.5`
o3	OpenAI	`o3`
grok45	xAI	`grok-4.5`
mistral_large	Mistral	`mistral-large-2512`
deepseek_v4	DeepSeek, via Together	`deepseek-ai/DeepSeek-V4-Pro`
minimax	MiniMax, via Together	`MiniMaxAI/MiniMax-M3`
kimi	Moonshot, via Together	`moonshotai/Kimi-K2.6`
inkling	Thinking Machines, via Together	`thinkingmachines/Inkling`
llama33	Meta, via Together	`meta-llama/Llama-3.3-70B-Instruct-Turbo`
command_a (outside panel)	Cohere	`command-a-plus-05-2026`
gemini3pro (outside panel)	Google	`gemini-3.1-pro-preview`

Request. One user message carrying the scenario, both questions and their options in bank order, and the allocation instruction; the framing, where there is one, as the system prompt. The options are in the same order on every call. A request seed of 1000 plus the rerun index goes to OpenAI, xAI, Together (`seed`) and Mistral (`random_seed`); no seed was sent to Anthropic, Google or Cohere. Token ceilings are per provider: Anthropic 3072, OpenAI and xAI 4096, Together and Google 6144, Mistral and Cohere 2048. We did not set temperature, top-p, top-k or reasoning effort; seeds and output ceilings were configured as listed, and each model ran at its provider's defaults for the rest, which the run records do not capture.

Scenarios. Forty-eight, twelve per axis, each with a judgment question and a reasoning question; 93 of the 96 questions have three options and 3 have four. Twelve scenarios, three per axis, carry the human baseline (b12). Eight framings per scenario. Each (model, framing) cell is one file of 240 responses (48 scenarios times 5 reruns). No cell was run more than once. The build scripts read `runs/` non-recursively and raise if a duplicate complete cell appears; a superseded run would be moved to `runs/_superseded/`, which they do not read.

Exclusions. A response whose point allocations could not be parsed from the model's reply is marked `extraction_failed` and excluded from every score; the runner's uniform fallback allocation is never scored. A file of 240 responses is therefore not a cell of 240 scored answers. The run records carry no finish-reason field, so a truncated reply is not distinguishable from a malformed one after the fact. Across the panel 205 of 21,120 responses are excluded, concentrated in Kimi (99) and MiniMax (53); the largest single loss is Kimi's geometry cell, 31 of 240 (3 of the 60 on the baseline twelve). Command A's 176 of 720 include 101 failed calls.

Columns are the eight framings in the order above (neutral, individualist, collectivist, hierarchical, egalitarian, seasonal, geometry, color), then the total excluded over the responses in the file.

model	neu.	ind.	col.	hie.	ega.	sea.	geo.	clr.	excluded / responses
deepseek_v4	0	0	0	0	0	0	0	0	0 / 1920
gpt55	0	0	0	0	0	0	0	0	0 / 1920
grok45	0	0	0	0	0	1	0	3	4 / 1920
inkling	5	0	1	0	1	0	0	0	7 / 1920
kimi	3	14	9	10	13	6	31	13	99 / 1920
llama33	0	0	0	0	0	1	0	0	1 / 1920
minimax	5	6	5	3	8	13	9	4	53 / 1920
mistral_large	4	0	16	4	0	0	0	0	24 / 1920
o3	2	2	1	1	1	4	5	1	17 / 1920
opus	0	0	0	0	0	0	0	0	0 / 1920

model	neu.	ind.	col.	hie.	ega.	sea.	geo.	clr.	excluded / responses
sonnet	0	0	0	0	0	0	0	0	0 / 1920
command_a (outside panel)	38	79	59	-	-	-	-	-	176 / 720
gemini3pro (outside panel)	0	-	-	-	-	-	-	-	0 / 240

Human respondents. Sixty-eight, a convenience sample recruited within one to two degrees of the author, demographically varied but concentrated in educational and professional background (almost all technology consultants and professionals), scored with the identical function. Their item exposure is not the models': 58 respondents, on instrument version 1.1, answered four scenarios, one per axis drawn by rotation from that axis's three baseline scenarios; ten respondents, on version 1.0, answered all twelve. Most human axis scores are therefore single-item observations, and each baseline item has between 19 and 35 human responses. Consent and licence terms are in `DATA-LICENSE.md` and decision 6.

A2. The scoring function

Every axis score, for humans and models alike, comes from one function (`compute_dimensional_score` in `scenario_bank.py`). Each scenario belongs to exactly one axis and poses two questions, judgment (what should happen) and reasoning (why it matters), each with three or four answer options. Every option carries a fixed signed loading toward its axis, a value in the interval from -1 to +1: an option that fully marks one pole loads at +1 or -1, an option that leans partway loads at a fraction (the bank uses +0.3, -0.3, +0.5, -0.5 and +0.7), and a neutral option loads at 0.

A respondent distributes a fixed pool of points across the options for each question. The judgment score is the loading-weighted average of the points they placed:

j = sum over judgment options of (loading times points), divided by the sum of points placed

The reasoning score r is the same over the reasoning options. Because points are non-negative and the loadings lie in the interval from -1 to +1, both j and r fall in that same interval: all points on full +1 options gives +1, all on full -1 options gives -1, points on neutral options pull toward 0. An even split across a question's options scores the mean of its loadings, which is zero only where the loadings are balanced; in 24 of the 48 scenarios an even split across both questions scores away from zero (between -0.15 and +0.17), or one question has no option at a full pole (mac_1's reasoning options load -1, 0 and -0.5, so its attainable maximum is +0.6). The same options are scored for humans and models, which standardises the scoring rule and does not establish measurement equivalence: the two differ in how they place points (A8) and in item exposure (A3), and either can interact with unbalanced loadings. The axis score combines the two,

score = 0.6 times j, plus 0.4 times r,

weighting the judgment question above the reasoning question. A position on an axis pools every scored response on that axis's scenarios, each response's allocation normalised to sum to one, so each scored response carries equal weight; a model's position is not a mean of five rerun-level scores, although with no exclusions the two coincide. The blank-question rule applies at that pooled level: if no points were placed on one question across the pooled responses, the other is used at full weight rather than deflating the score toward zero. The 0.6 to 0.4 split is the single free parameter in the scoring; A8 shows the compression holds under judgment-only, reasoning-only, and the combined weighting, so the finding does not depend on it.

A3. Compression

Dispersion is population standard deviation; the model figures are each model's unframed neutral position (A5 shows the panel widening two to five times under framing). The N-matched draw takes 11 of the 68 human scores on the axis at random without replacement, 100,000 times, and counts how often that subsample is at least as tight as the 11-model panel. It describes the observed human sample; it is not a test of equal dispersion between two populations. The ratio CI is a 5,000-draw percentile bootstrap resampling humans and models independently.

Axis	Human SD	Model SD (neutral)	Ratio (human/ model)	95% CI on ratio	draws as tight as the panel
Moral Agent	0.412	0.061	6.78x	[4.94, 13.38]	0 of 100,000
Authority	0.401	0.052	7.69x	[5.82, 12.63]	0 of 100,000
Moral Domain	0.327	0.061	5.37x	[3.66, 11.73]	0 of 100,000
Obligation Scope	0.361	0.064	5.65x	[4.59, 8.49]	0 of 100,000

Zero of 100,000 draws bounds the tail at about three in 100,000 (one-sided 95 percent), not at one in 100,000. The lower bound of every ratio CI is at least 3.66. The human sample is a convenience sample concentrated among technology professionals (A1).

Item exposure. The human SD above is mostly the spread of single-item scores across people (A1); the model SD is the spread of positions each pooling three items and up to five reruns. The two exposures can be matched two ways.

On the ten respondents who answered all twelve scenarios:

Axis	Human SD (n = 10)	Model SD	Ratio
Moral Agent	0.348	0.061	5.72x
Authority	0.192	0.052	3.69x
Moral Domain	0.406	0.061	6.67x
Obligation Scope	0.309	0.064	4.84x

Item by item, the humans who answered that item against the eleven models' positions on it, reruns still pooled on the model side:

item	axis	humans	human SD	model SD	ratio
mac_1	Moral Agent	33	0.404	0.111	3.64x
mac_2	Moral Agent	27	0.338	0.048	7.11x
mac_3	Moral Agent	28	0.457	0.114	4.02x
auth_1	Authority	34	0.319	0.067	4.78x
auth_2	Authority	35	0.464	0.070	6.60x
auth_3	Authority	19	0.340	0.057	5.97x
domain_1	Moral Domain	34	0.382	0.090	4.24x
domain_2	Moral Domain	21	0.351	0.042	8.40x
domain_3	Moral Domain	33	0.386	0.117	3.30x
scope_1	Obligation Scope	34	0.277	0.048	5.78x
scope_2	Obligation Scope	29	0.433	0.126	3.43x
scope_3	Obligation Scope	25	0.352	0.107	3.29x

The model SD is the smaller on every item, by 3.3 to 8.4 times. The ten-respondent subset is small and on a different instrument version; the item rows still pool model reruns.

Rerun averaging. Each model position pools up to five reruns where each human score is one response. Two approximations of what a single model draw would show. `validity/aggregation_artifact.py` inflates the observed panel variance by the square of the median within-model run SD times 0.8 (one minus one over five), an illustrative variance-inflation calculation that treats the pooled position as an equal-weight mean of five runs: the four ratios become 6.32, 7.25, 5.03 and 5.14, and for any ratio to reach 3 under the same additive-noise arithmetic, within-person noise would have to be 64 to 83 percent of the observed human variance, a hypothetical figure, not a measured human reliability. At those inflated model SDs the N-matched draw gives 0, 0, 0 and 1 of 100,000 (seed

20260911). The second approximation selects one observed rerun per model at random, each rerun still pooling the three baseline items, and takes the human SD over the panel SD of the selection, 20,000 selections, seed 20260911:

Axis	Median ratio across selections	Central 95 percent of selections
Moral Agent	6.17	[4.92, 8.34]
Authority	6.97	[5.16, 9.99]
Moral Domain	5.02	[4.45, 5.85]
Obligation Scope	4.99	[3.85, 7.35]

These describe how the ratio moves with which rerun is chosen, not a population interval. Neither approximation makes a model's three-item axis score equivalent to a one-item human score.

Where the two populations sit, for context (the compression claim is about spread, not location):

Axis	Human mean	Human median	Model mean
Moral Agent	+0.29	+0.36	-0.05
Authority	+0.17	+0.20	+0.12
Moral Domain	+0.37	+0.43	+0.29
Obligation Scope	+0.37	+0.42	+0.28

On Moral Agent the model mean and the human center sit on opposite sides of the midpoint.

A4. Run reliability (test-retest)

Within-model dispersion is the standard deviation of a cell's axis score across its five reruns, reported as the median over the eleven models. Between-model dispersion is the panel SD from A3. The spread between models is 1.9 to 2.5 times the spread a single model shows on rerun. This within-model dispersion is what the interactive viewer draws as its optional run-spread overlay. No sampling temperature was sent; each model ran at its provider's default, which the run records do not capture, so the within-model figures are not on a common footing across models.

Axis	Within-model run SD (median)	Between-model SD	Between / within
Moral Agent	0.026	0.061	2.33
Authority	0.021	0.052	2.50
Moral Domain	0.025	0.061	2.41
Obligation Scope	0.033	0.064	1.95

A5. Frame responsiveness

Displacement is the mean absolute per-axis change from a model's own neutral position, averaged over the four axes; the group figures average that across models and framings. The intervals are 20,000-draw percentile bootstraps that resample the eleven models, each model carrying all its framings, so a model's framed displacements, which share its unframed baseline, stay together; the difference and the ratio are computed on the same resampled models. They describe how far the panel figure moves when the observed eleven are reweighted, conditional on these framings, and are not confidence intervals for models in general.

Quantity	Estimate	95% model-resampling interval
Cultural (4 framings)	0.363	[0.302, 0.429]
Nonsense (geometry, color)	0.203	[0.150, 0.259]
Seasonal, non-moral (not a clean null)	0.246	[0.197, 0.298]
Cultural minus nonsense	0.160	[0.129, 0.199]
Nonsense over cultural	0.558	[0.464, 0.641]

Per framing:

Framing	Mean displacement
individualist	0.335
collectivist	0.442
hierarchical	0.543
egalitarian	0.132
irrelevant (seasonal)	0.246
nonsense: geometry	0.195
nonsense: color	0.211

The seasonal framing (labeled irrelevant in the data) was designed as a non-moral noise floor. Its displacement (0.246) is nearer the nonsense framings than zero, and its interval overlaps theirs and not the cultural framings'; it is reported as a non-moral framing, not a control.

The cultural result rests on direction, not distance. Each cultural framing names a principle that maps to one axis, so the four framings target two of the four axes. For each, the signed shift on its target axis in the expected direction, and the count of models that moved that way:

Framing	Target axis	Mean signed shift (expected direction)	Models in expected direction
individualist	Moral Agent (toward autonomous)	+0.46	11 / 11
collectivist	Moral Agent (toward relational)	+0.56	11 / 11
egalitarian	Authority (toward skeptical)	+0.16	11 / 11
hierarchical	Authority (toward deferential)	+0.62	11 / 11

Under both nonsense framings all eleven models move the same way on Moral Agent (toward relational: geometry mean -0.25, color mean -0.24, 0 of 11 moving the other way). The nonsense prompts share the cultural prompts' scaffold, a society with social roles and moral obligations, so the shared shift is not attributable to geometry or color as against that scaffold; the pilot has no scaffold-only framing.

Per model, displacement under the two nonsense framings as a share of displacement under the four cultural framings:

Model	Nonsense / cultural
sonnet	0.27
kimi	0.39
opus	0.46
llama33	0.47
o3	0.53
minimax	0.53
mistral_large	0.54
inkling	0.62
deepseek_v4	0.72
gpt55	0.72
grok45	0.73

Kimi's geometry cell is scored on 209 of 240 responses (A1).

Between-model dispersion. The compression in A3 is measured at neutral. Under framing the panel does not stay equally tight: the standard deviation across the eleven models, averaged over the four axes, rises two to five times above its neutral value under every framing, cultural or nonsense alike.

Framing	Between-model SD (b12)	vs neutral	Between-model SD (all 48)	vs neutral
neutral	0.059	1.0x	0.033	1.0x
individualist	0.125	2.1x	0.129	3.9x
collectivist	0.148	2.5x	0.132	4.0x
hierarchical	0.160	2.7x	0.154	4.7x
egalitarian	0.130	2.2x	0.074	2.2x
irrelevant (seasonal)	0.129	2.2x	0.093	2.8x
nonsense: geometry	0.145	2.4x	0.097	2.9x
nonsense: color	0.142	2.4x	0.109	3.3x

The nonsense and seasonal framings widen the panel about as much as the cultural ones. What differs under the cultural framings is the shared direction of the signed shifts above, not the spread.

A6. Cross-lab clustering

Each model is fingerprinted by its deviation from the panel consensus, and its nearest neighbor is the model with the highest Pearson correlation of deviations, a pattern-similarity measure rather than a distance. We report two grains. Seven of the eleven models are their lab's only model, so their nearest neighbor is cross-lab by construction; only the two Anthropic and the two OpenAI models have a same-lab candidate, and under label exchange the expected number of same-lab nearest-neighbor links is 0.4. The counts below are the observed pattern, not evidence that lab has no association with position.

Panel grain (deviation of the four axis positions, all 48 scenarios): no model's nearest neighbor is a same-lab sibling (0 of 11). A correlation over four values has two degrees of freedom; the values run high and carry little.

Model	Nearest neighbor	r	Same lab
deepseek_v4	opus	0.882	no
gpt55	mistral_large	0.995	no
grok45	minimax	0.971	no
inkling	llama33	0.889	no
kimi	gpt55	0.717	no
llama33	inkling	0.889	no
minimax	sonnet	0.984	no

Model	Nearest neighbor	r	Same lab
mistral_large	gpt55	0.995	no
o3	llama33	0.739	no
opus	deepseek_v4	0.882	no
sonnet	minimax	0.984	no

Fine grain (deviation of the 48 per-scenario positions): ten of eleven nearest neighbors are cross-lab. The single exception is Opus, whose nearest neighbor is Sonnet at r = 0.137, narrowly ahead of two OpenAI models (gpt55 at 0.115, o3 at 0.104). Sonnet itself is not the mirror of that tie: its own nearest neighbor is MiniMax at r = 0.500, a much stronger and cross-lab pairing. The same-lab pairing appears at the scenario grain only.

Model	Nearest neighbor	r	Same lab
deepseek_v4	o3	0.223	no
gpt55	inkling	0.144	no
grok45	sonnet	0.394	no
inkling	gpt55	0.144	no
kimi	gpt55	0.112	no
llama33	o3	0.415	no
minimax	sonnet	0.500	no
mistral_large	o3	0.233	no
o3	llama33	0.415	no
opus	sonnet	0.137	yes
sonnet	minimax	0.500	no

The tightest cross-lab pairs are MiniMax and Sonnet at 0.50, which crosses country as well as company, and o3 and Llama at 0.415, two American labs.

A7. Reasoning token spend

Reasoning tokens are whatever the provider's usage object reports for the call; no reasoning or thinking setting was sent (A1). OpenAI, xAI and Together return a `reasoning_tokens` count, Google a thoughts count. The Anthropic, Cohere and Mistral adapters record no such field, and Together returns none for Llama, so four panel models have no reported count. That is a property of the request and the provider's accounting, not a count of zero, and it says nothing about whether those models reasoned.

Providers also differ on whether reasoning tokens sit inside the output count, so no share of output is given. Mean reported reasoning tokens per unframed scenario, with each model's Euclidean distance from the panel centroid across the four axes (all 48 scenarios):

Model	Reasoning tokens / scenario	Responses with a count	Distance from panel center
kimi	3628	240	0.031
minimax	1189	240	0.058
inkling	1141	235	0.085
deepseek_v4	1066	240	0.058
grok45	494	240	0.093
o3	247	240	0.055
gpt55	131	240	0.060
llama33	not reported	0	0.061
mistral_large	not reported	0	0.109
opus	not reported	0	0.042
sonnet	not reported	0	0.079

Among the seven models with a count, spend runs from 131 to 3,628. Their distances from the panel center run from 0.031 to 0.093 and do not order with spend: Kimi, with the largest reported count, is the closest of all eleven, and GPT-5.5, with the smallest, sits at 0.060. Seven models with different providers, accounting and ceilings support exploratory description only; no correlation is reported. The contrast in the report, Kimi at 3,628 against GPT-5.5 at 131, is two rows of this table. Reasoning-token spend, as reported, does not order position on this instrument; nothing here tests reasoning in any other setting, and the counts confound model, provider accounting, reasoning mode and token ceiling.

A8. Sensitivity analyses

Scope. Model SD at neutral on the 12 baseline scenarios versus all 48. The panel is tighter on the full set, as expected from averaging over more items; the panel's concentration does not depend on the subset. The humans answered the twelve only, so the full set carries no human comparison.

Axis	Model SD (b12)	Model SD (all 48)
Moral Agent	0.061	0.031
Authority	0.052	0.028
Moral Domain	0.061	0.052
Obligation Scope	0.064	0.023

Question weighting. Model SD under judgment-only, reasoning-only, and the combined default stays in the same narrow band on every axis:

Weighting	Moral Agent	Authority	Moral Domain	Obligation Scope
judgment only	0.070	0.045	0.072	0.065
reasoning only	0.059	0.069	0.050	0.075
combined (0.6 / 0.4)	0.061	0.052	0.061	0.064

With the human side re-scored under the same weighting, the human-to-model ratio:

Weighting	Moral Agent	Authority	Moral Domain	Obligation Scope
judgment only	6.93x	9.19x	5.20x	6.70x
reasoning only	8.10x	7.35x	7.95x	4.95x
combined (0.6 / 0.4)	6.78x	7.69x	5.37x	5.65x

Range coverage. On all 48 scenarios, the span of the eleven unframed positions as a share of each axis's fixed range of 2:

Axis	Lowest model	Highest model	Span	Share of range
Moral Agent	-0.068	+0.044	0.112	5.6 percent
Authority	+0.046	+0.126	0.080	4.0 percent
Moral Domain	-0.052	+0.106	0.158	7.9 percent
Obligation Scope	+0.117	+0.203	0.086	4.3 percent

This is a min-to-max span of model positions, a different quantity from the SD ratios in A3, and the humans did not answer these 48 scenarios, so it carries no human comparison.

Panel composition. Adding back the two models outside the panel (13 models) leaves the panel SD small on every axis (Moral Agent 0.076, Authority 0.061, Moral Domain 0.061, Obligation Scope 0.077). Command A's unframed cell is scored on 202 of 240 responses (A1).

Allocation style. The models and the humans place points differently. On the baseline twelve, unframed, the share of points on neutral options is similar, but on average the models spread their points across options far more evenly than the humans: mean normalised allocation entropy 0.85 to 0.95 per model against 0.46 to 0.51 for the humans, and a largest option holding about half the points against about two thirds. The figures are means over responses; the human rows weight a twelve-item respondent three times a four-item one, and the equal-person mean of the humans' judgment entropy is 0.48. Judgment / reasoning:

Respondent	Neutral-option share	Full-pole share	Normalised entropy	Largest option's share
humans (352 responses)	0.15 / 0.20	0.56 / 0.53	0.46 / 0.51	0.66 / 0.61
deepseek_v4	0.22 / 0.19	0.50 / 0.53	0.85 / 0.88	0.55 / 0.49
gpt55	0.21 / 0.19	0.51 / 0.54	0.88 / 0.91	0.54 / 0.47
grok45	0.20 / 0.18	0.56 / 0.59	0.91 / 0.94	0.49 / 0.43
inkling	0.23 / 0.24	0.48 / 0.49	0.87 / 0.89	0.54 / 0.50
kimi	0.21 / 0.18	0.51 / 0.55	0.90 / 0.93	0.50 / 0.45
llama33	0.18 / 0.18	0.58 / 0.59	0.94 / 0.95	0.45 / 0.41
minimax	0.21 / 0.21	0.53 / 0.54	0.93 / 0.95	0.49 / 0.43
mistral_large	0.17 / 0.20	0.62 / 0.57	0.93 / 0.92	0.49 / 0.45
o3	0.20 / 0.20	0.54 / 0.55	0.93 / 0.94	0.48 / 0.43
opus	0.23 / 0.17	0.49 / 0.58	0.89 / 0.93	0.52 / 0.44
sonnet	0.23 / 0.21	0.52 / 0.53	0.92 / 0.95	0.49 / 0.44

A flatter allocation scores nearer the mean of its options' loadings. As a sensitivity to the allocation format, every allocation on both sides is collapsed onto its largest option. Ties at the maximum are common, 43 and 40 of the 352 human allocations and 70 and 95 of the 655 model ones (judgment and reasoning; 22 of Llama's 60 for each), so the rule for them matters; the primary rule shares the points equally among tied maxima, which does not privilege option order, and the first-tied and last-tied rules are shown beside it.

Axis	Human SD	Model SD	Ratio, ties shared	Ratio as scored
Moral Agent	0.604	0.265	2.27x	6.78x
Authority	0.582	0.119	4.89x	7.69x
Moral Domain	0.490	0.154	3.18x	5.37x
Obligation Scope	0.467	0.111	4.21x	5.65x

Tie rule	Moral Agent	Authority	Moral Domain	Obligation Scope
shared among tied maxima	2.27x	4.89x	3.18x	4.21x
first tied option	2.35x	3.84x	3.19x	4.03x
last tied option	2.13x	6.20x	3.10x	4.37x

Item by item under the shared rule, the humans who answered the item against the eleven models on it:

item	ratio	item	ratio	item	ratio	item	ratio
mac_1	1.23x	auth_1	3.01x	domain_1	4.70x	scope_1	6.11x
mac_2	2.95x	auth_2	2.45x	domain_2	2.95x	scope_2	1.56x
mac_3	1.19x	auth_3	2.70x	domain_3	1.41x	scope_3	2.28x

The model spread stays the smaller on every axis under every rule and on every item, and the ratios change substantially with the scoring rule: 2.3 to 4.9 with ties shared, 2.1 to 6.2 across the three rules, 1.2 to 6.1 item by item. Collapsing is a change of measurement, not a decomposition: it discards secondary preferences, magnifies differences near ties and changes human as well as model scores, so it shows the ratio's sensitivity to allocation format and does not isolate how much of the as-scored ratio is response style. Model reruns are still pooled here.

Exclusions. Scoring the excluded responses with whatever allocation the parser stored, a uniform one where no object parsed and a padded or truncated one where the option count was wrong, instead of dropping them, moves the b12 model SD by at most 0.0011 (Obligation Scope, 0.0639 to 0.0650; Moral Domain 0.0609 to 0.0605; Authority 0.0521 to 0.0522; Moral Agent unchanged), with a largest per-model shift of 0.027; Kimi's displacement under the geometry framing is 0.223 with its 31 excluded responses dropped and 0.196 with them scored. A stored allocation is not the answer the model failed to give, and a uniform one pulls toward the loading mean, so this bounds the arithmetic of the exclusion, not its cause.

A9. Validity

Three results bear on validity.

Known-groups. The human sample sits autonomous, skeptical, narrow and universal (A3). The bank's global sign convention (decision 1) was checked against that orientation when it was fixed (`analysis/README.md`), so the profile's orientation is not independent evidence; the sample is a convenience sample from the author's network (A1).

Response to manipulation. The models move under each cultural framing in the expected direction on its target axis, all eleven on all four (A5); under both nonsense framings all eleven move the same way on Moral Agent. The cultural prompts prescribe the principle they are named for, so this shows the models follow a prescribed perspective; it does not show that a framed position matches any population.

Reliability. Within-model run-to-run SD is roughly half the between-model SD on every axis (A4), under an identical prompt on every rerun (A1).

Not established: the factor structure and inter-item consistency of the bank; convergent validity against an independent instrument, which can be assessed within one population; measurement invariance across cultural groups, which is required before any cross-cultural comparison and needs cross-cultural human samples on this instrument; and criterion validity against behavior, whether a position here relates to what a model does in open-ended use, which needs a behavioral study.

A10. Reproducibility

Three stdlib scripts regenerate everything: `build_figures.py` (compression, bootstrap, panel clustering, the three figures), `build_viewer_data.py` (the viewer payload, including per-cell run dispersion), and `build_appendix.py` (this appendix's `appendix_stats.json`, including the reasoning-token and between-model-dispersion tables). A fourth, `build_csv.py`, exports the tidy CSV tables. Seeds by script: `build_figures.py` 20260719; `build_appendix.py` 20260720 for the A3 draws, 20260910 for the A5 model resampling, 20260911 for the rerun selection and the recount. Human response files are read in sorted filename order. The scenario bank with every option's loading (`scenarios.json`), every framing prompt (`framings.json`), the run files with every response and allocation (`runs/`), the anonymised human responses and their licence (`human-responses/`, `DATA-LICENSE.md`) and the decision log are in the public repository, github.com/DeclanMichaels/reasoner-pilot. Every random draw is seeded as listed. Each script raises on a duplicate complete cell so a re-run cannot silently change a number.

Analysis is stdlib-reproducible from the raw runs. Responsibility for the work, and for any errors in it, is mine alone. Methodology was AI-assisted and that assistance is disclosed.

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